

# Rakey | Ruoqing Yang

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## EDUCATION

- **University of Nottingham** 2023–2024  
MSc, Human Computer Interaction (graduated with Distinction) *Nottingham, United Kingdom*
- **Ningbo University** 2018–2022  
BSc, Logistics Management *Ningbo, China*
- **University of Oxford** 2022  
Winter School, Artificial Intelligence and Machine Learning *Oxford, United Kingdom*

## ACADEMIC PAPERS

1. **R. Yang** and C. Greenhalgh. “Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli.” *Virtual Reality*, in press, 2026.
2. S. Jin, **R. Yang**, W. Tong, and L. H. Lee. “AIs or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants.” *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1155–1156, 2026.
3. S. Jin, **R. Yang**, and L. H. Lee. “Who Should Serve You in Metaverse Luxury Stores? A Comparative Study of 3D Avatars, AI Agents, Webcam Staff, and Voice Assistants.” *Electronic Commerce Research and Applications*, under review, 2026.

## ACADEMIC EXPERIENCE

- **Comparing VR and AR for Dual-Phobia Virtual Exposure Therapy** 2024–2025  
*MSc Dissertation, Individual Research · University of Nottingham* *Nottingham, United Kingdom*
  - Built the same exposure scenarios in both VR and AR to compare the two media across spider and contamination phobia, following a research-through-design approach.
  - Conducted a mixed-methods study (n=31) with questionnaires, behavioural metrics, and semi-structured interviews, analysing the data through two-way repeated-measures ANOVA and reflexive thematic analysis.
  - Found VR elicited higher experience intensity overall, while AR performed comparably and was more ecologically valid in the spider scenario, pointing to a staged AR-then-VR design.
- **User Perception of Different Shopping Assistants in VR Retail** 2025  
*Research Collaboration · University of Cambridge* *Cambridge, United Kingdom*
  - Contributed to the design of a study comparing 4 shopping-assistant modes (3D avatar, webcam, voice, AI) in a luxury VR retail setting, defined the experimental conditions and participant tasks.
  - Co-built the VR retail scenes in Unity and conducted the user study (n=20), running participant sessions and collecting and cleaning the multi-source perceptual and behavioural data.
  - Found 3D avatars to be the optimal assistant mode, while identifying a fundamental realism-immersion trade-off in asymmetric PC-VR interaction.

## PROFESSIONAL EXPERIENCE

- **AI Video Specialist** 01/2024–07/2024  
*O Shaped · Internship* *London, United Kingdom*
  - Built AI avatar and generative-video workflows for educational content, cutting production time by 40%.
  - Iterated on the generated outputs from user feedback, improving clarity and usability under real production constraints.
  - Explored digital-human and embodied avatar formats for more present, human-like content delivery.

## PROJECTS

- **One’s Own Room – AI-Customised WebXR Safe Zone** [\[Github\]](#)
  - Designed a WebXR app: a user-generated, interactive 3D room for real-time AI dialogue, giving users full control over emotional disclosure.
  - Engineered the system pipeline integrating WorldLabs’ custom 3D Gaussian splatting scenes, Three.js/Spark shaders, and real-time conversational voice interactions driven by ElevenLabs.
- **Virtual Exposure Therapy Application** [\[Github\]](#)
  - Built a Unity VR/AR exposure-therapy prototype for Meta Quest 3, presenting spider and contamination scenarios across AR and VR conditions.
  - Used passthrough to switch between AR and VR, with hand-tracking for controller-free interaction. Users can touch a static spider to trigger its crawling state.
- **AR Gallery Treasure Hunt Game** [\[Video\]](#)
  - Developed exhibition environment setups and modelled 3D content based on UoN Lakeside Arts exhibit materials and venue geography.
  - Designed location-based AR for spatial exploration, achieving 90% task completion rate.

## SKILLS & INTEREST

**Research Methods:** Research-through-Design, Mixed Methods (Quant/Qual)   **Data Analysis:** SPSS, Python  
**XR / Game Dev:** Unity (C#), WebXR (Three.js, WebGL), Spark AR   **Design:** Adobe Suite (Ps/Ai/Ae), Figma, Blender  
**Languages:** English, Mandarin, Taiwanese Hokkien   **Interest:** AI Art (AI Shrine – GOSIM Spotlight, Paris 2026)